Observe, the following examples.

$100 = 10 \times 10 = 10^2$

This can also be expressed as the product of two different bases with the same exponent as, $100 = 25 \times 4 = (5 \times 5) \times (2 \times 2) = 5^2 \times 2^2$

We notice that 5 and 2 are the bases and 2 is the exponent.

In the same way, $a \times a \times a \times b \times b = a^3 \times b^2$

Think



Justify your answer.

Fig. 3.2

Consider, $35 = 7^1 \times 5^1$, where there is no repetition of factors. Thus, usually $7^1 \times 5^1$ is represented as 7×5. So, when the power is 1 the exponent will not be explicitly mentioned.

Note

1. The exponents 2 and 3 have special names 'squared' and 'cubed' respectively. For example, 4^2 is read as 'four squared' and 4^3 is read as 'four cubed'.
2. The other name of exponent is indices. Do you remember the word 'indices'? In Class VI, we heard of this word while applying BIDMAS rule in simplification. For example,

$6^3 + 4 \times 3 - 5 = (6 \times 6 \times 6) + 4 \times 3 - 5$ [BIDMAS]
$= 216 + (4 \times 3) - 5$ [BIDMAS]
$= 216 + 12 - 5$ [BIDMAS]
$= 228 - 5$ [BIDMAS]
$= 223$

Try these

Observe and complete the following table. First one is done for you.

Number	Expanded form	Exponential Form	Base	Exponent
216	6×6×6	6^3	6	3
144		12^2		
	(–5)×(–5)		–5	
		m^5		
343			7	3
15625	25×25×25			

Example 3.1 Express 729 in exponential form.

3	729
3	243
3	81
3	27
3	9
	3

Solution

Dividing by 3, we get

$729 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^6$

Also, $729 = 9 \times 9 \times 9 = 9^3$

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